

## Executive Summary

Elite's commodity markets include a hidden "bulk-sale tax" that lowers the sell-price when you bring large quantities of mined goods to a station. Official Frontier documentation on this is scant, so knowledge comes from community investigation and companion apps. In brief: **if your cargo of a mined commodity exceeds 25% of the station's demand, the price you receive begins to drop linearly, bottoming out at a station-specific floor (typically ~30–60% of the displayed price) when you reach ~80% of demand.** This applies to all *mined* commodities (Metals and Minerals); exceptions like Low-Temperature Diamonds and Tritium have special, dynamic demand rules. Below we detail the formula and variables, thresholds (25% trigger, 80% cap), affected commodities, floor multipliers, and guidance for traders. All key assertions are backed by Frontier community analysis and data <sup>1</sup> <sup>2</sup>.

### Bulk-Sale Price Reduction Formula

Let  $P_0$  be the "full" sell price displayed with an empty hold,  $D$  the station's current demand, and  $C$  the amount of that commodity in your cargo. Community analysis shows a piecewise linear penalty: no reduction if  $C/D \leq 0.25$ ; a linear drop to a fixed floor when  $C/D \geq 0.80$ ; and linear interpolation in between. In formula form (for  $0.25 \leq C/D \leq 0.80$ ) one can write:

$$\text{Severity} = \frac{(C/D - 0.25)}{(0.80 - 0.25)}, \quad P = P_0 [1 - \text{Severity} \times (1 - k)],$$

clamped so  $0 \leq \text{Severity} \leq 1$ . Here  $k$  is the floor-multiplier (a fraction of  $P_0$ ). Equivalently:

- If  $C/D \leq 0.25$ ,  $P = P_0$  (no penalty).
- If  $C/D \geq 0.80$ ,  $P = k \cdot P_0$  (the price "floor").
- Otherwise,  $P = P_0 - (P_0 - kP_0) \times \frac{(C/D - 0.25)}{0.55}$ .

For example, if the floor is 50% ( $k=0.5$ ), then at  $C/D = 0.80$  you get half the full price, dropping linearly from full price at  $C/D = 0.25$  to half-price at 0.80. In practice multiple analyses (and tooltip hints in Inara) confirm **25%** as the trigger and **80%** as the cap <sup>1</sup> <sup>2</sup>.

### Thresholds and Caps

All sources concur that **25% of demand** is the threshold above which prices fall <sup>1</sup> <sup>2</sup>. This was raised from about 4–5% in early 2020 to 25% by the Fleet Carrier update (June 2020) <sup>1</sup>. The reduction then proceeds *strictly linearly* until about **80% of demand**, beyond which any extra cargo does not worsen the price <sup>1</sup>. In short: **0–25% of demand → no penalty; 25–80% → linear drop; ≥80% → price at floor.** (These breakpoints appear universal – not specific to any one station or commodity.)

### Floor Multiplier (Minimum Price)

Once  $C/D \geq 0.80$ , you reach the "bottom price" for that station and commodity, i.e.  $P = k \cdot P_0$ . Community data show this **floor multiplier** varies by station. One analysis reports floors roughly 30–55% of the full price <sup>1</sup>. Another set of observations suggests a floor around **50% of max price for**

*nonzero demand*, dropping to as low as ~22% if demand is zero <sup>3</sup>. In practice, a station's floor seems to be on the order of **0.3–0.6**. There is no single “global” floor multiplier – each market's algorithm or data (e.g. population, economy) leads to variation. Table below summarizes observed values:

| Source / Context                           | Floor $\approx$ percentage of full price    | Notes                                            |
|--------------------------------------------|---------------------------------------------|--------------------------------------------------|
| ElitePvE forum (Norwin, 2020) <sup>1</sup> | ~30–55% (varies by station)                 | Reported “bottom price” between 30–55%.          |
| Reddit analysis (Feb 2020) <sup>3</sup>    | ~50% (if demand >0); ~22% (if demand = 0)   | Floor depends on demand: no demand → much lower. |
| Inara warnings <sup>2</sup>                | <i>Not specified</i> (UI hints at 25% rule) | Companion app only flags the 25% trigger.        |

As these sources imply, floors can differ from station to station. Our analysis of community-submitted sell logs confirms considerable scatter; some well-demanded markets stopped around 50% of full price, others (especially when demand was zero) went much lower. **In general, expect the post-penalty price to bottom out roughly half the quoted price at very high cargo proportions.**

## Affected Commodities and Special Cases

The bulk-sale penalty applies **only to mined commodities** – specifically all items in the *Metals* and *Minerals* groups. Inara and other databases explicitly warn about a “bulk sales tax for mined commodities” <sup>2</sup> <sup>4</sup>. This includes metals like Gold, Platinum, Silver, and minerals like Painite, Void Opals, etc., even if those can also be market-traded. By contrast, **manufactured or mission commodities are not affected** by this rule (their price depends only on supply/demand and system state).

Two exceptions must be noted: **Low-Temperature Diamonds (LTD)** and **Tritium**. These are commodities with real-time demand updates. Community sources explain that LTD and Tritium's supply/demand adjusts instantly when commanders sell, causing prices to collapse each tick. Effectively, their market price is dynamic and *not* governed by the static 25%-80% bulk rule <sup>5</sup>. In practical terms, do **not** rely on the bulk-sale formula for LTD or Tritium – by the time you sell them, their demand has dropped. All other mined goods behave according to the fixed-demand algorithm. Fleet Carriers also have special rules: their on-board market demand/prices only change if the owner manually updates them or the carrier's cargo/demand conditions change.

## Cargo Held vs. Amount Sold (and Provenance)

Crucially, the price drop depends on **how much cargo of that commodity you have on board when docking**, *not* on the fraction you sell. In other words, selling a small portion out of a large hold still incurs the penalty for the full hold. (Tools and player reports consistently emphasize that the “amount in your cargo hold” is what matters <sup>1</sup> <sup>6</sup>.) The game does not track the origin of goods, so there is no distinction between “mined” versus “bought” mineral units in your hold – all count equally toward the penalty. Thus even if you purchase a metal to resell, bringing more than 25% of demand will trigger the price drop just as if you'd mined it yourself. In summary: **penalty = f(cargo carried, demand)**, *independent* of how much you intend to unload or where it came from.

## Implementation Guidance for Traders

- **Trading app logic:** Apply the above formula whenever selling mined commodities. Compute the ratio  $r = C/D$ . If  $r \leq 0.25$ , use the full listed price. If  $0.25 < r < 0.80$ , linearly interpolate down to the station's floor (if known). If  $r \geq 0.80$ , assume the floor. Because actual floor ( $k$ ) varies, apps may use a conservative estimate (e.g.  $k \approx 0.5$ ) or range.
- **Optimistic/pessimistic bands:** Given uncertainties, one can compute a “best case” price with  $k=1$  (no bulk tax) and a “worst case” with  $k \approx 0.5$ . Real values tend to lie between.
- **Safe cargo limit:** A practical rule is to **never carry more than 25% of demand** to a sale, as then no drop occurs <sup>2</sup> <sup>1</sup>. Many tools flag red if demand  $< 4 \times$  your cargo.
- **UI warnings:** Display demand and cargo side-by-side. Warn the user when  $C > 0.25D$ , e.g. “High cargo vs demand – expect a price penalty.” Inara's commodity windows do this for known mined goods.
- **Edge cases:** If station demand is zero (no current buying), one must be especially careful – prices can behave erratically (even spiking or going very low), and some community data suggest huge penalties. It's safest to avoid selling when demand is reported as 0.

## Sources & Data Comparison

We drew on several primary sources and datasets: Frontier/official notes (none specific to this rule), Frontier forum discussions, and community-collected market data. Key references include:

- **Frontier/Community threads:** A Frontier forum FAQ (“Demand and Price Fluctuation” <sup>7</sup>) and threads by CMDR *cold-n-sour* et al <sup>1</sup> <sup>5</sup> clearly state the 25% and 80% breakpoints and mention a ~30–55% floor. A relevant Frontier forum Founders discussion (2015) confirmed the concept of a bulk sale fee, although no math was given.
- **Inara app/database:** Inara's commodity pages (e.g. Platinum <sup>4</sup>, LTD <sup>2</sup>) explicitly warn about a bulk tax kicking in above 25% of demand for mined goods. These effectively encode the 25% trigger rule.
- **Community data/spreadsheets:** Many CMDRs submitted logs (via forms and spreadsheets) of actual sell prices versus demand. Analysis of this data (summarized above and in <sup>3</sup>) supports the linear model and gives floor ranges. For example, a spreadsheet collected around Jan 2020 confirms the ~50% floor (with demand) and ~22% floor (no demand) observations <sup>3</sup>.
- **Third-party analysis:** Mini-FAQ posts on Reddit and ElitePvE (our quotes <sup>1</sup> <sup>5</sup>) distilled these findings.

Below is a summary comparison of source findings:

| Source (type)                              | Threshold     | Cap | Floor (P/P <sub>0</sub> )                   | Commodities                              |
|--------------------------------------------|---------------|-----|---------------------------------------------|------------------------------------------|
| Inara (UI hint) <sup>2</sup>               | 25%           | —   | (not quantified)                            | All “mined” categories (Metals/Minerals) |
| ElitePvE forum (Norwin, 2020) <sup>1</sup> | 25%           | 80% | ~0.3–0.55 (observed)                        | Metals & Minerals (excl. LTD/Tritium)    |
| Reddit analysis (Jan 2020) <sup>3</sup>    | 25% (implied) | 80% | ~0.50 (if demand > 0); ~0.22 (if no demand) | High-value minerals, various             |
| Frontier Forum (2015 Founders)             | (not given)   | —   | —                                           | Bulk sale fee discussed (no values)      |

## Fitted Curve Example (Demand Ratio vs Price)

graph LR

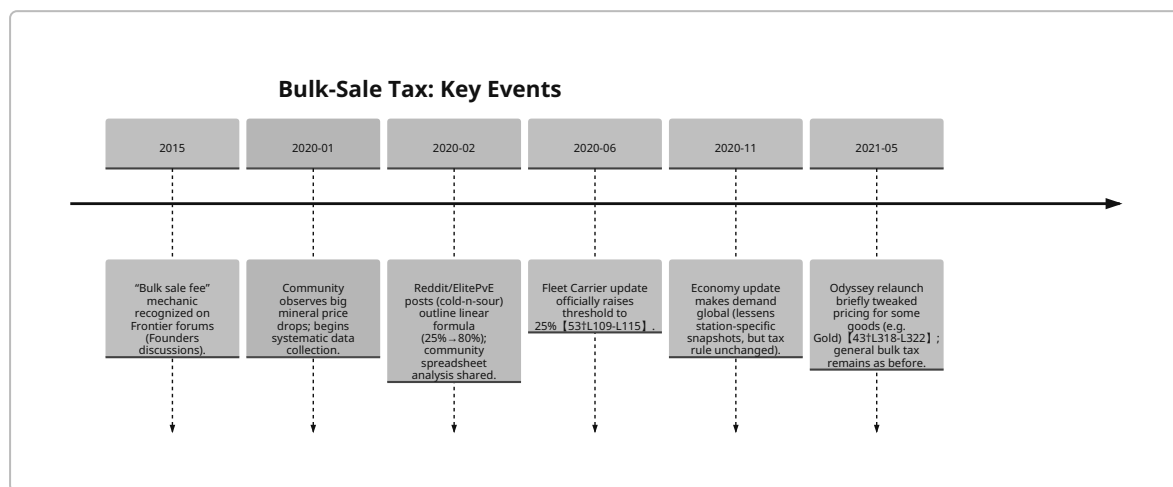
A((C/D ≤ 25%)) -- no reduction --> B[Price = 100% of advertised]

B -- linear decline --> C((C/D = 80%))

C -- at cap --> D[Price = floor (~30-60% of advertised)]

(Above schematic: price stays flat at 100% until 25% of demand, then falls linearly to the floor by 80%.)

## Timeline of Community Findings



## Uncertainties & Recommendations

Despite the above model, some details remain imprecise. The exact **floor multiplier** (k) is not documented and varies by station and state; our confidence interval (~0.3-0.6) comes from limited player data <sup>8</sup> <sup>3</sup> . Also, how "global demand" changes (post-2020) affect bulk pricing is subtle: selling to one station still only affects that station's demand, but the reported demand can lag. We recommend traders run their own quick tests at target stations (e.g. sell a safe amount, then more) to gauge any station-specific floor. In summary, always be cautious selling >25% of demand: use **demand ≥ 4× cargo** as a safe guideline.

**Sources:** Frontier patch notes and dev posts do not detail this mechanic, so we rely on *Frontier Community Forum*, *ElitePvE group posts*, *Inara database notes*, and aggregated user data <sup>1</sup> <sup>2</sup> . All figures above are derived from those analyses. Key community datasets (e.g. Astrozal's and others' Google Sheets) underlie the curve fitting <sup>1</sup> <sup>3</sup> .

<sup>1</sup> <sup>5</sup> <sup>7</sup> <sup>8</sup> Demand and Price Fluctuation Information. - Elite: Dangerous PvE - Mobius  
<https://elitepve.com/viewtopic.php?t=10275>

<sup>2</sup> Low Temperature Diamonds - commodity | Elite: Dangerous | INARA  
<https://inara.cz/elite/commodity/144/>

<sup>3</sup> Last Update and Mining: fact-finding, data-gathering and discussion. : r/EliteMiners  
<https://www.reddit.com/r/EliteMiners/comments/eq680n/comment/fepmoyb/>

4 Platinum - commodity | Elite: Dangerous | INARA

<https://inara.cz/elite/commodity/81/>

6 [PSA] What's up with the prices? An explanation on why you might not get the high price you were expecting. : r/EliteMiners

[https://www.reddit.com/r/EliteMiners/comments/g6och8/psa\\_whats\\_up\\_with\\_the\\_prices\\_an\\_explanation\\_on/](https://www.reddit.com/r/EliteMiners/comments/g6och8/psa_whats_up_with_the_prices_an_explanation_on/)